Writing I2C Clients in LINUX

I2C is a protocol for communication between devices. In this column, the author takes the reader through the process of writing I2C clients in Linux.

I2C is a multi-master synchronous serial communication protocol for devices. All devices have addresses through which they communicate with each other. The I2C protocol has three versions with different communication speeds – 100kHz, 400kHz and 3.4MHz. The I2C protocol has a bus arbitration procedure through which the master is decided on the bus, and then the master supplies the clock for the system and reads and writes data on the bus. The device that is communicating with the master is the slave device.

The Linux I2C subsystem

The Linux I2C subsystem is the interface through which the system running Linux can interact with devices connected on the system's I2C bus. It is designed in such a manner that the system running Linux is always the I2C master. It consists of the following subsections.

I2C adapter: There can be multiple I2C buses on the board, so each bus on the system is represented in Linux using the *struct i2c_adapter* (defined in *include/linux/i2c.h*). The following are the important fields present in this structure.

bus number: Each bus in the system is assigned a number that is present in the I2C adapter structure which represents it.

I2C algorithm: Each I2C bus operates with a certain protocol for communicating between devices. The algorithm that the bus uses is defined by this field. There are currently three algorithms for the I2C bus, which are pca, pcf and bitbanging. These algorithms are used to communicate with devices when the driver requests to write or read data from the device.

I2C client: Each device that is connected to the I2C bus on the system is represented using the *struct i2c_client* (defined in *include/linux/i2c.h*). The following are the important fields present in this structure.

Address: This field consists of the address of the device on the bus. This address is used by the driver to communicate with the device.

Name: This field is the name of the device which is used to match the driver with the device.

Interrupt number: This is the number of the interrupt line of the device.

I2C adapter: This is the *struct i2c_adapter* which represents the bus on which this device is connected. Whenever the driver makes requests to write or read from the bus, this field is used to identify the bus on which this transaction is to be done and also which algorithm should be used to communicate with the device.

I2C driver: For each device on the system, there should be a driver that controls it. For the I2C device, the corresponding driver is represented by *struct i2c_driver* (defined in *include/linux/i2c.h*). The following are the important fields defined in this structure.

Driver.name: This is the name of the driver that is used to match the I2C device on the system with the driver.

Probe: This is the function pointer to the driver's probe routine, which is called when the device and driver are both